

One CSI-Derived Rate Cannot Serve Every Budget: Budget Overrun and the Age Horizon in 5G Link Adaptation

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Abstract—A CSI report drives one rate decision, but different consumers hold that decision to different error budgets. We measure the resulting mismatch on an NR LDPC-based link-level simulation, in two ways. First, budget overrun. A selection policy whose achieved BLER sits near an eMBB budget of 10^{-1} overruns a URLLC budget of 10^{-3} on the same channel by a factor of 112, and the overrun persists at every SNR from cell edge to cell centre; a budget-aware policy removes it with explicitly specified protection. Second, age. With the link calibrated to actually meet the 10^{-1} budget when CSI is fresh, we ask how long a report may be held before the budget is lost. The answer is a handful of milliseconds at walking speeds and a single transmission interval at moderate mobility and above, so there periodic reporting meets the budget only at the frame-minimum interval, and HARQ-driven correction is necessary for any less aggressive reporting cadence. Every number comes from a sealed, pre-registered record, including a superseded relaxed-threshold pilot that this version recomputes at the calibrated target.

Index Terms—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

I. INTRODUCTION

Adaptive modulation and coding turns a CSI report into a rate decision. The transmitter selects the highest MCS whose required SINR the report clears, and the selection is tuned to an error budget. The trouble is that the traffic behind one report does not share one budget. An eMBB flow tolerates a block error rate of 10^{-1} ; a URLLC flow tolerates 10^{-3} or less. The same decision, from the same report, on the same channel, is judged against both.

This paper measures that mismatch with two quantities that stay fixed throughout. The first is the achieved BLER $\hat{B}$ of a selection policy. The second is the budget overrun $\Omega(\beta) = \hat{B}/\beta$, the achieved BLER in units of the consumer's budget. The overrun is where the consumer-relativity lives: $\hat{B}$ is a property of the policy and the channel, while Ω depends on who is asking.

Two results follow. The first (§III) is partly mechanical and we say so: a policy run near 10^{-1} must overrun 10^{-3} by about a hundredfold. The measured content is the exact overrun (112), its persistence across the whole SNR range including the cell edge, and the quantified cost of the budget-aware policy that removes it. The second (§IV) is the paper's core. With the link calibrated so that fresh CSI genuinely meets

the 10^{-1} budget, the age horizon, the longest a report can be held before the budget is lost, collapses to a few transmission intervals at 10 Hz Doppler and to a single interval above 50 Hz. An earlier exploration at a relaxed threshold suggested a coherence-proportional law; the calibrated recomputation shows that at the budget itself there is almost no horizon to scale. The corrections are old, outer-loop link adaptation above all, and our measurement says when they have stopped being optional.

II. METHOD

Substrate. We use an NR LDPC-based link-to-system simulation. For each MCS of the NR table (TS 38.214 Table 5.1.3.1-1; QPSK-64-QAM, code rates $\geq 1/5$) we measure a block-error curve $e_m(\gamma)$ with the Sionna 5G LDPC decoder [2] over AWGN, 400 decoded blocks per point on a -8 to 28 dB grid at 2 dB spacing, linearly interpolated between points. The time-domain link draws a per-TTI SNR trace from a 3GPP TR38.901 TDL-A channel (3.5 GHz, 30 ns delay spread) and resolves each transmission as $\text{NACK}(t) \sim \text{Bernoulli}(e_m(\gamma(t)))$. Two accuracy limits follow and bound our claims. At 400 blocks per point, a measured zero errors still leaves a binomial 95% upper bound near 7.5×10^{-3} , so the curves are trustworthy at the 10^{-2} level and above, and any smaller value is an interpolation, not a count; the linear interpolation itself is a disclosed simplification that is coarse on steep tails. And the SNR-to-error mapping ignores fading within a TTI. Table I lists the parameters.

System model. Let $\gamma(t)$ be the true SNR in dB at TTI t and $\hat{g}(t)$ the estimate the transmitter holds. Fresh CSI reads $\hat{g}(t) = \gamma(t) + n(t)$ with $n \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 1$ dB. A report held with period P reads $\hat{g}(t) = \gamma(t_k) + n(t_k)$ with $t_k = P\lfloor t/P \rfloor$: the estimate carries the report noise of its issue time and then ages. The certificate selects, at reference target $\beta_0 = 10^{-1}$ with backoff Δ ,

$$m^*(\hat{g}) = \max\{m : \rho_m(\beta_0) + \Delta \leq \hat{g}\}, \quad (1)$$

where $\rho_m(\beta) = \min\{\gamma : e_m(\gamma) \leq \beta\}$. The achieved BLER is estimated two ways: the empirical NACK rate

$$\hat{B} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{NACK}(t)] \quad (2)$$

TABLE I
LINK-LEVEL PARAMETERS

Carrier / TTI	3.5 GHz / 1 ms
Channel model	3GPP TR38.901 TDL-A, DS = 30 ns
Mean SNR	12 dB
Decoder curves	Sionna 5G NR LDPC, 400 blocks/pt, -8-28 dB @ 2 dB
MCS set	NR TS 38.214 Tbl 5.1.3.1-1 (QPSK-64QAM)
CQI report noise σ	1 dB
Trace length	6000 TTI (budget); 12 000 TTI (age)
eMBB / URLLC budget	10^{-1} / 10^{-3}
Budget-aware policy	$\Delta = 4$ dB + $K=3$ independent branches
OLLA steps	$\delta_{\text{up}} = 0.1$ dB, $\delta_{\text{dn}} = 0.9$ dB
Age sweep	$f_D = 10\text{--}400$ Hz; $P = 1\text{--}32$ TTI; calibrated Δ_{cal}

for the age study, and the expected rate $\hat{B} = \frac{1}{T} \sum_t e_{m^*}(\gamma(t))$ for the budget study, whose 10^{-3} scale $T=6000$ cannot resolve by counting. The consumer-relative observable is the *budget overrun* $\Omega(\beta) = \hat{B}/\beta$. $\hat{B}$ itself does not depend on β , which is why the eMBB and URLLC-naive columns of Table II(a) coincide by construction; the two consumers differ in Ω : the same $\hat{B} \approx 0.112$ is $\Omega \approx 1.1$ for eMBB and $\Omega \approx 112$ for URLLC.

Budget-aware policy. It raises the backoff to $\Delta = 4$ dB and combines $K=3$ diversity branches, independent TDL-A realizations from distinct seeds standing in for frequency or spatial replicas: a block fails only if all branches fail. A backoff alone cannot reach 10^{-3} , because a single Rayleigh branch spends about 4% of its time in fades too deep for any MCS. Independence makes the diversity a best case, and the claim rests on a product, not a directly counted deep tail: the per-branch failure rates sit at the percent level, where the curves are reliable, and their independence product lies below the 10^{-3} budget with two orders of margin. The absolute product value ($\approx 10^{-5}$) inherits the interpolation's error and should be read as an order of magnitude.

OLLA. The correction of §IV selects $m^*(\hat{g}(t) + o(t))$ and updates the offset on HARQ feedback,

$$o(t+1) = o(t) + \delta_{\text{up}} \mathbf{1}[\text{ACK}(t)] - \delta_{\text{dn}} \mathbf{1}[\text{NACK}(t)], \quad (3)$$

with $\delta_{\text{up}} = 0.1$ dB and $\delta_{\text{dn}} = \delta_{\text{up}}(1-\beta)/\beta$, so ACKs push the selection up and the loop settles where the NACK rate equals β . This recurrence matches the verified implementation line for line.

The age horizon. Because the naive fresh policy lands near but above the budget ($\hat{B} \approx 0.113$), the age study first calibrates: a fixed backoff Δ_{cal} , chosen in 0.25 dB steps on a disclosed calibration seed, brings the fresh policy to $\hat{B} \leq 0.09$ at every Doppler. All policies then carry Δ_{cal} . The age horizon is $P^*(f_D) = \max\{P : \hat{B}(P) \leq 0.10\}$; a Doppler at which no period qualifies is recorded as censored, never clamped.

Discipline. Each experiment is pre-registered with numerical bars fixed before the graded run; records are content-hashed and graded on seeds disjoint from design; failed, voided and superseded outcomes stay in the record. Review of an earlier draft found that the age sweep had used a relaxed 0.15 threshold while the text claimed 0.10; that exploration is kept, and §IV reports the calibrated recomputation.

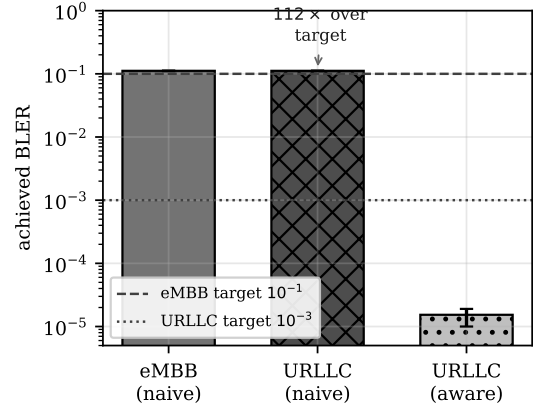


Fig. 1. One report, two budgets (NR LDPC link-to-system; 3 sealed seeds, bars are means, whiskers min-max). The same selection sits near the eMBB budget and overruns the URLLC budget by 112 $\times$; the budget-aware policy meets it.

III. THE RELIABILITY BUDGET

A base station serving mixed traffic derives one CSI estimate per user and picks an MCS from it. Consider the same estimate judged under two budgets, 10^{-1} for eMBB and 10^{-3} for URLLC. Figure 1 shows the sealed result, three graded seeds of 6000 TTI each with fresh CSI. The naive policy achieves $\hat{B} \approx 0.112$, near its eMBB budget. The URLLC consumer reads the same 0.112 as an overrun of 112 $\times$. Part of this is arithmetic, and we do not claim otherwise: a policy run near one budget must overrun a budget a hundred times stricter by about that factor. What the measurement adds is the exact figure, the fact that nothing in the scheduler's view flags it, and a feasible policy that removes it: the budget-aware policy of §II meets the 10^{-3} budget with margin. Its protection is explicitly specified, a 4 dB backoff and threefold diversity, though we do not quantify the resulting spectral-efficiency loss here.

The overrun is not an artifact of the operating point. A separately sealed cell sweeps the mean SNR from 3 to 21 dB (Fig. 2). The certificate rides the eMBB budget, so the URLLC overrun stays between 77 $\times$ and 142 $\times$ at every in-range SNR and is still 54 $\times$ where the MCS table saturates. Below about 6 dB even the eMBB consumer loses its budget, since the lowest MCS is already too fast for the channel ($\hat{B} \approx 0.33$ at 3 dB). No margin repairs a certificate whose weakest format is too aggressive.

IV. THE AGE HORIZON

The second axis is time, and it is where the budget question bites hardest. A report is fresh when issued and ages as the channel decorrelates. At 200 Hz Doppler with a 20-TTI report period, the held estimate is off by 5.7 dB on average (mean absolute error; aging dominates, report noise contributes about 1 dB), and the naive certificate runs at $\hat{B} \approx 0.36$ against the 0.10 budget, an overrun of 3.6 $\times$ from age alone. OLLA, Eq. (3), corrects the same held report and holds 0.10 (three

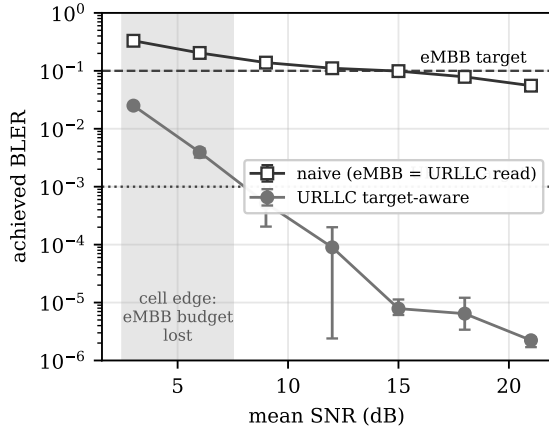


Fig. 2. The overrun across mean SNR (sealed sweep, 3 seeds, whiskers min-max). The URLLC overrun is roughly a hundredfold at every in-range SNR; below about 6 dB (shaded) even the eMBB budget is lost.

sealed seeds, 0.34–0.37 naive against 0.101–0.102 corrected; the per-seed record is in the public ledger [7]).

How long may a report be held before the budget is lost? An exploratory sweep at a relaxed 0.15 threshold suggested a coherence-proportional law, $P^* \approx 0.18 T_{\text{coh}}$; it was superseded by the calibrated 0.10 experiment reported here, and both records are kept. The calibrated recomputation at the true 0.10 budget gives a starker answer (Fig. 3, Table II(b)). With the fresh baseline calibrated to $\hat{B} \leq 0.09$ at every Doppler (a 0.5 dB backoff suffices), the horizon over the three graded seeds is $P^* = 4$ TTI at 10 Hz (6 on one seed), 2 at 25 Hz (3 on one seed), and 1 TTI, the frame minimum, at 50 Hz and above, with no censored points. There is no room for a proportionality law because there is almost no horizon to scale: once the budget is tight, the first few milliseconds of aging consume it. At 3.5 GHz these Dopplers are modest speeds: 10 Hz is about 3 km/h, a walk, and 50 Hz about 15 km/h, a bicycle. The operational statement follows. Below walking pace a report survives a few TTIs; at moderate mobility and above, periodic reporting meets the budget only at the frame-minimum interval, so HARQ-driven correction is necessary for any less aggressive reporting cadence. The coherence-proportional behaviour is real but belongs to relaxed thresholds, where degradation has room to accumulate before the budget is lost.

V. RELATED WORK

Link adaptation, CSI aging and outer-loop correction are old subjects. OLLA [1] holds an error target under imperfect or stale CSI by adjusting an SINR offset on HARQ feedback; it is exactly the correction our age result makes mandatory. Short-packet analysis [3] establishes the reliability regime where the budget gap lives. NR configures CSI reporting as periodic, semi-persistent, or DCI-triggered aperiodic [4]; none of these adapt the cadence to the consumer’s budget, and our horizon result says how little cadence alone can do at a tight one. Event-based sampling such as send-on-delta [5]

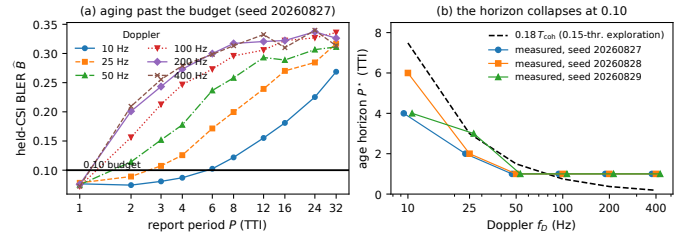


Fig. 3. The age horizon at the true 0.10 budget (calibrated baseline; three sealed graded seeds). (a) Held-CSI BLER against report period. (b) The horizon collapses to a few TTI at low mobility and the frame minimum above 50 Hz.

TABLE II
SEALED PER-SEED RESULTS (ACHIEVED BLER $\hat{B}$)

(a) Budget study (§III), fresh CSI				
seed	eMBB	URLLC-naive	URLLC-aware	
20260824	0.1119	0.1119	1.7×10^{-5}	
20260825	0.1103	0.1103	1.9×10^{-5}	
20260826	0.1125	0.1125	$< 10^{-5}$	
(b) Age horizon (§IV), calibrated $\Delta_{\text{cal}} = 0.5$ dB				
seed	$P^*(10)$	$P^*(25)$	$P^*(\geq 50)$	fresh worst
20260827	4	2	1	0.080
20260828	6	2	1	0.078
20260829	4	3	1	0.080

Seed IDs are campaign date-stamps; (a) expected rate, (b) empirical NACK rate. In (a) the first two columns are the same trace: $\hat{B}$ does not depend on the budget; the consumers differ in $\Omega = \hat{B}/\beta$ (1.1 vs 112). In (b) P^* is in TTI; “fresh worst” is the largest fresh-CSI $\hat{B}$ over Dopplers, confirming the calibrated baseline meets the budget. No Doppler was censored.

is a natural extension we tested only in exploratory form (a sealed confirmation attempt was voided by a control check and we claim nothing). Learned CSI feedback trained on reconstruction error [6] is active; in our controlled tests the objective gap it might suggest did not appear at practical operating points. We offer no new estimator, scheduler or offset rule. We offer the measurement: achieved BLER against the consumer’s budget, and the age horizon at the budget itself.

VI. DISCUSSION AND CONCLUSION

The unit of correctness in link adaptation is not the CSI report. It is the pair of report and budget. Sized to the consumer’s budget, margins and reporting cadence keep the certificate honest; sized to a nominal channel, they do not, and the costs are now numbers: a $112\times$ overrun for a mismatched budget, and an age horizon of a few milliseconds at a tight one. The evidence has limits worth stating. It is an NR LDPC-based link-to-system simulation, one TDL-A profile, one MCS table topping at 64-QAM, and deep-tail values below 10^{-3} are curve-resolution-limited; over-the-air HARQ traces are the natural next step. In companion sealed experiments, two margin policies with equal fleet budgets received opposite verdicts from two user fleets whose failures ride different axes of the channel (cells XPROTO-CSI-FLIP and XPROTO-QOT-FLIP in the sealed repository [7]); a full treatment of that inversion, with scheduler interpretations and goodput, is beyond this

paper's scope and is reported separately. The methodology is not radio-specific; the same budget-relative failure appears in optical quality-of-transmission and in database replication [7]. The results here stand on the 5G NR certificate alone.

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